NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

Quick-Trial Lab mainline

Group Quick-Trial Lab

Status Promoted

Period 2026-10-15 23:46 — 2026-10-17 11:40

Stage conclusion

Primary choice is K-Dense, with Academic as a style reference; covers styling and export only—layout must be hand-coded and architecture manually verified

Project report

Redrawing the Graph-DeepONet Final Network Architecture Diagram with Scientific Plotting Skills

Objective

Verify the repositories and licenses of 6 candidate Skills (AgentFigureGallery, Nature Skill, Academic Figure Skill, Origin Style Replot, figures4papers, K-Dense Scientific Visualization), select at most 2 (prioritizing those that output editable files) to test-draw the "final network architecture diagram" of the openclaw urban wind-field Graph-DeepONet project, and render a worth-using / drop verdict.

How the Test Target Was Selected

Candidates: the 04-09 figure (UNet Speed Distillation Round 34, Global Best; boxes and connector lines overlap), checked against later final versions in the same directory: `UNet_7level_final.drawio` (04-15) and the project notes (recording the best model as HybridGraphCNN+FineGraph48+Fourier64, with no existing diagram). Which was chosen and why: both the round34 original figure (04-09) and `UNet_7level_final.drawio` (04-15) belong to a separate UNet reproduction track (`auto_v2`, Lu et al. 2025) and do not depict Graph-DeepONet. The optimal architecture of Graph-DeepONet exists only in text notes (04-02), where lines 35–36 specify a HybridGraphCNN + FineGraph48 + Fourier64 trunk; therefore a schematic was drawn from the text record, with any unrecorded layer counts or channel dimensions explicitly labeled "unrecorded" (original text verified by the Lead on 10-16). To test the specific question raised by Liang Hu—"can the overlaps in the original figure be fixed?"—a supplementary test was also run redrawing the round34 original figure.

Current Status

10-16 03:20 Wrap-up: Worth using (with limitations), per the headline conclusion. Graph-DeepONet has no existing final architecture diagram, only text records; the schematic drawn from those records is labeled "not verified against code" and cannot be used as a final journal submission figure.

Security Audit

See the audit report. Lead correction: `eval(parse(...))` at `compose.py:419` in Academic appears only inside a docstring usage example and is never executed anywhere in the repository (verified via `grep`), representing low risk and not grounds for disqualification.

Trials and Results

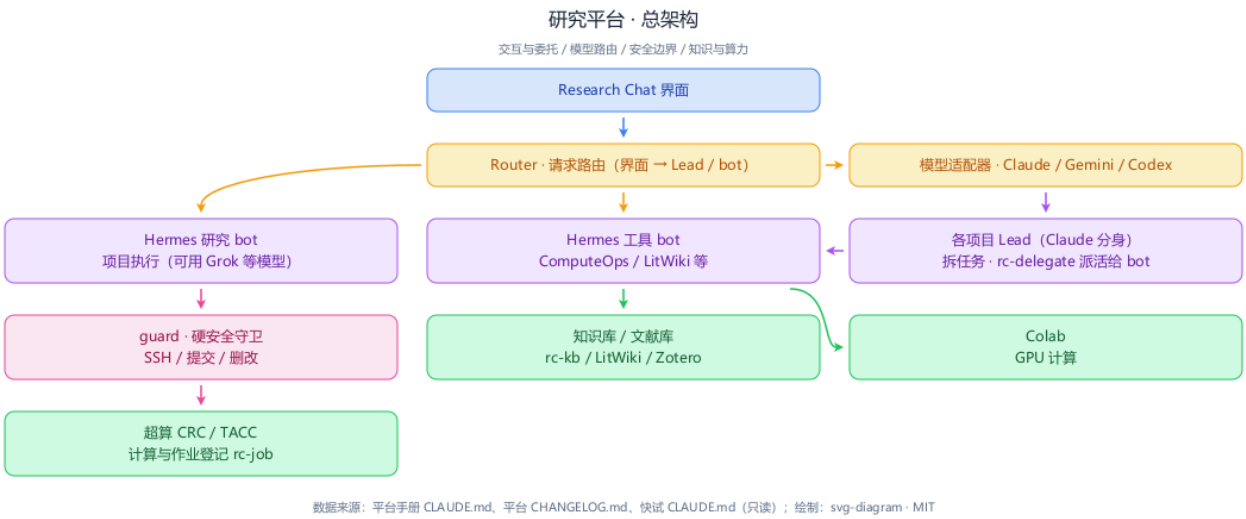
Date	Action Taken	One-Line Result
10-16	Supplementary test: redrew the round34 original figure with both Skills	Verified by Lead: title, subtitle, numbers across all 9 modules, 3 skip connections, and auxiliary head all match the original figure; all box-on-box overlaps and curve-through-box crossings in the original are eliminated; each SVG has 38 editable text elements and the PDFs have embedded fonts
10-16	Audited 6 repositories: 5 located, no matching repository found for Origin Style Replot; figures4papers is CC BY-NC	Selected Academic Figure Skill (Apache-2.0) and K-Dense `scientific-visualization` (MIT)

Date	Action Taken	One-Line Result
10-16	Drew Graph-DeepONet schematic from text records	Both figures are consistent in content, overlap-free, and have editable SVG text; the K-Dense version has excessive outer margins; all unrecorded parameters are explicitly labeled without fabricated numbers

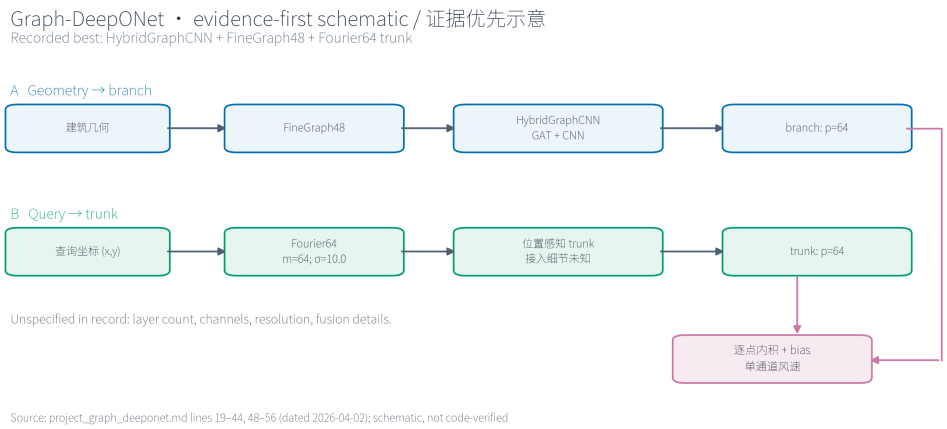
Open Items and Next Steps

- Hand implementation recommendation to Platform Lead: equip the plotting bot (`figures`) only with the `scientific-visualization` subdirectory of K-Dense pinned to commit `a1043ef544`, without installing the full 177-skill repository; from Academic, take only the style reference documents under `references`. Delivery gate: item-by-item checklist against the original figure/notes + visual PNG inspection + editable SVG text.
- If a final submission figure for Graph-DeepONet is needed: first obtain the source code or configuration of the best model to verify the exact layer counts and channel dimensions.

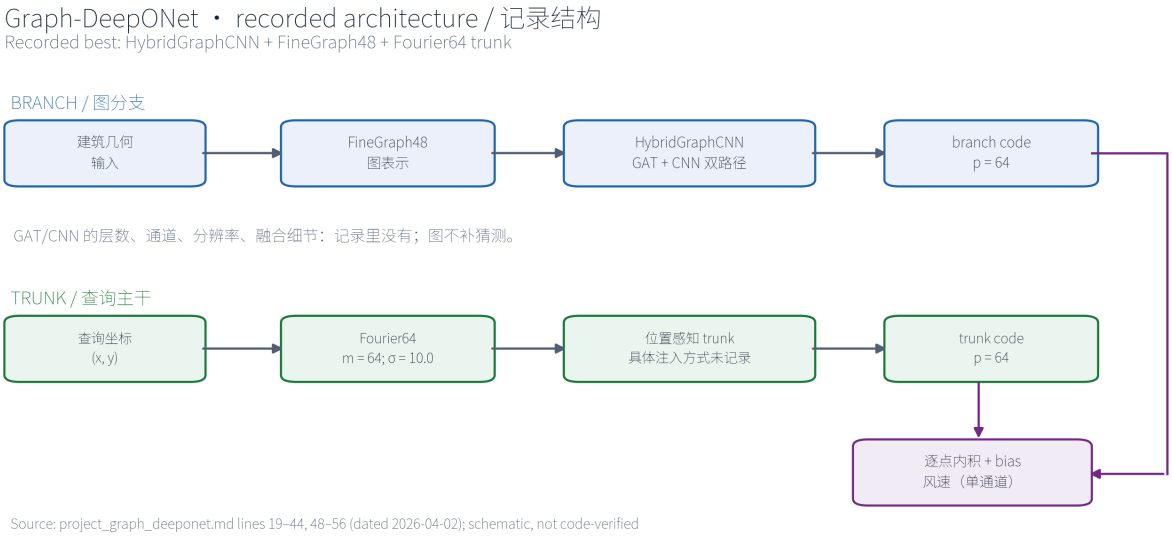
Key figures



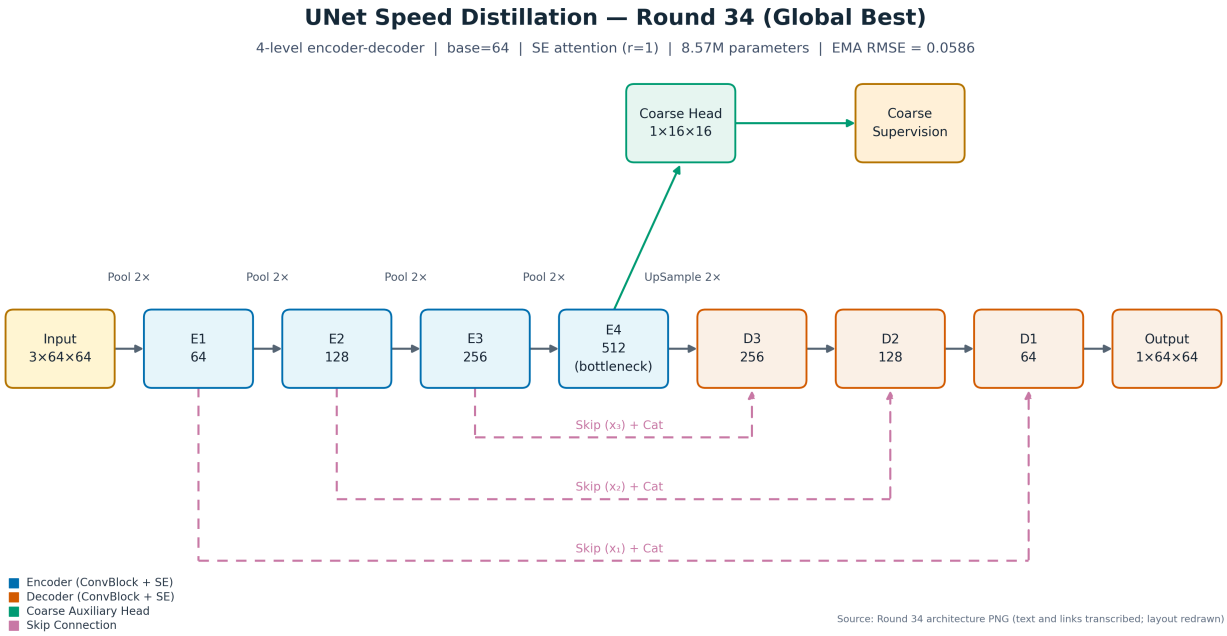
research platform architecture v4



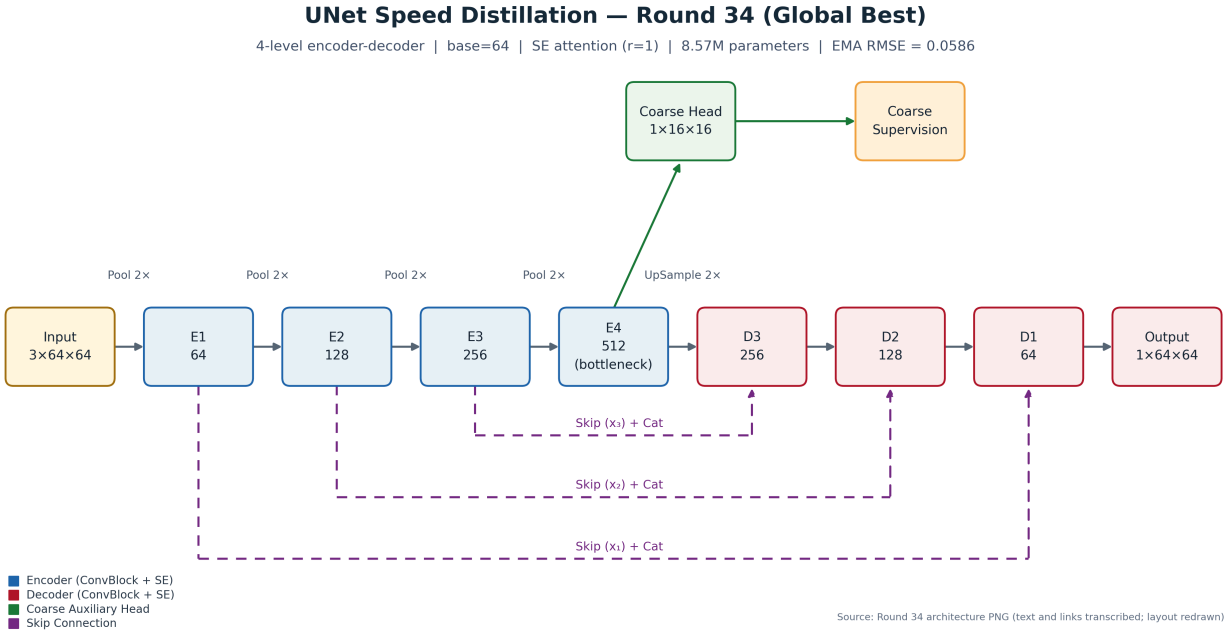
kdense visualization v3



academic figure v3



kdense round34 v2



academic round34 v2